

Pon Aung

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Portfolio: PonAppliedPhysics.github.io

EDUCATION

The Ohio State University

Columbus, OH

Bachelor of Science in Physics

Minor: Engineering Science, Mechanical Engineering Emphasis

Coursework: Statics, Mechanics of Materials, Dynamics, Material Science, Intermediate Electricity and Magnetism, Experimental Physics Instrumentation, Optics, AI for Mechanical Engineering (University of Michigan), Electronics

EXPERIENCE

Maval Industries

Twinsburg, OH

Mechanical Engineering Intern

June 2026 – Present

- Created and revised SolidWorks product prints and engineering drawings for power steering and gearbox assemblies, applying DFM/DFA principles to support remanufacturing processes
- Supported reverse engineering of electric power steering components by analyzing existing assemblies, documenting design features, and assisting with print creation for remanufacturing
- Conducted design of experiments to evaluate design and manufacturing process parameters, supporting optimization of machining, CNC, assembly, and remanufacturing workflows
- Collaborated with cross-functional engineering teams on tolerance analysis, mechanical design updates, and product development improvements for automotive steering systems
- Reported directly to the Senior Vice President of Operations, supporting engineering documentation, project tracking, and process improvement initiatives for steering component production

Honda Development and Manufacturing of America

Marysville, OH

Battery Safety Systems Project Lead – BEV Thermal Runaway Detection Capstone

August 2024 – May 2025

- Evaluated 9 thermal runaway detection technologies using weighted decision analysis to support mechanical design and thermal systems safety requirements for Honda BEV battery packs
- Designed ultrasonic pulse-echo experiments using 1 MHz transducers to characterize battery charge states; ANOVA confirmed statistically significant voltage variation during charging and discharging cycles
- Evaluated infrared pyrometer accuracy against FLIR thermal measurements at multiple distances, identifying surface emissivity effects that reduced detection reliability on reflective aluminum
- Validated HF gas sensor detection at 0.5, 1.5, and 2.5 m with coefficient of variation under 4 percent for timing and under 8 percent for concentration, confirming sub-1 ppm sensitivity at the longest test distance
- Delivered sensor validation data, selection matrix, and layered multi-sensor deployment strategy to Honda engineering leadership, completing all testing at 36 percent of the \$3,500 budget

PROJECT EXPERIENCE

OSU Underwater Robotics – Talos AUV

Columbus, OH

Mechanical Team Member

December 2025 – May 2026

- Developed CAD models and mechanical designs in SolidWorks for hull and chassis components; supported rapid prototyping and manufacturing processes using mills, lathes, and 3D printing to integrate a dual stereo-camera vision system into the Talos AUV for RoboSub
- Redesigned Delrin torpedo launcher assemblies through iterative design, tolerance analysis, and DFM principles, integrating mechanical seals and custom gearbox mechanisms that eliminated water-ingress failures from the prior RoboSub design
- Improved torpedo launcher accuracy and mechanical reliability by adding structural reinforcement and guided alignment features

TECHNICAL SKILLS

CAD & Manufacturing: SolidWorks, engineering drawings, product prints, DFM/DFA, tolerance analysis, reverse engineering, 3D printing, CNC process optimization

Programming & Data Analysis: Python, MATLAB, Excel, NumPy, SciPy, Pandas, Matplotlib, TensorFlow, data visualization, statistical analysis

Engineering Analysis: Quantitative analysis, mathematical modeling, Design of Experiments (DOE), computational modeling, heat transfer, uncertainty propagation, ANOVA

Lab & Testing: Experimental design, instrumentation and measurement, sensor validation, oscilloscope/DMM measurements, FLIR thermal imaging, ultrasonic testing

Technical Communication: Technical writing, engineering documentation, research presentations, cross-functional collaboration